

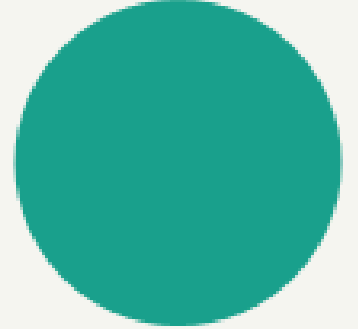
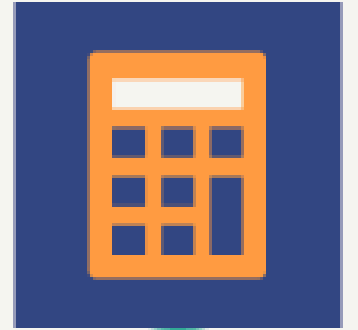
Try This!

Use the following frequency distribution for questions 1-3:

The frequency distribution table of the daily wages for a random sample of 20 workers in the drug industry is given as:

Class	15-	25-	35-	45-	55-	65-75	Total
Frequency(f)	6	2	5	3	2	2	20

- 1 The number of workers whose wages ranges from 25 up to 65 equals:
a) 13 (b) 12 (c) 15 (d) 16
- 2 The percentage of workers whose wages greater than or equal 35 is:
a) 60% (b) 65% (c) 85% (d) 80%
- 3 The mean wage equals:
a) 39.5 (b) 40.5 (c) 44 (d) 45.5
- 4 If the average depth of a lake is 1.5 meters, it means that
a) The deepest point of the lake is 1.5 meters.
b) An adult of average height can walk through the lake.
c) 50% of the lake has depth less than or equal to 1.5 meters.
d) There could be a spot in the lake where it is deeper than 1.5 meters.



Try This!

1

The following data set represents a random sample of the monthly income (in thousands of L.E) (40 , 3 ,3 ,4, 4, 6)

Find:

1) The range

2) The variance

3) The standard deviation

Choose :

2

If the statistics score of all students in some exam is 90 , this means that

The mode = zero b) The mode = 90

c) The Range= 90 d) The standard deviation = zero

3

In a data set, if all values are negative, then:

a)All measures of central tendency are negative, and all measures of dispersion are negative.

b)All measures of central tendency are positive, and all measures of dispersion are negative.

c)All measures of central tendency are negative, while all measures of dispersion are non-negative.

d)None of the above.

4

If $\text{Range}(\text{depth of an artificial lake}) = 0$ and $\text{median}(\text{depth}) = 1$ meter, this implies that

a)mean (depth) = 0

b)Standard deviation (depth) = 1

c)The sum of the deviations from 1 equals zero

d)Mode (depth) = 0

Titanic Dataset Analysis

Part 1 — Measures of Central Tendency & Dispersion

Task:

1. Load the Titanic dataset (seaborn has it built-in).
2. Pick one numeric column (e.g., age or fare).
3. Calculate: Mean, Median, Mode, Range, Variance (population & sample), Standard deviation (population & sample)

Part 2 — Create three different types of graphs using any attribute:

Task:

1. Bar Chart → Number of passengers per class (pclass)
2. Histogram → Age distribution (age)
3. Box Plot → Fare by passenger class (fare vs pclass)

P.S

- Write one paragraph explaining the meaning of each measure calculated.
- Write one paragraph interpreting each visual (e.g., "Most passengers were in 3rd class").